

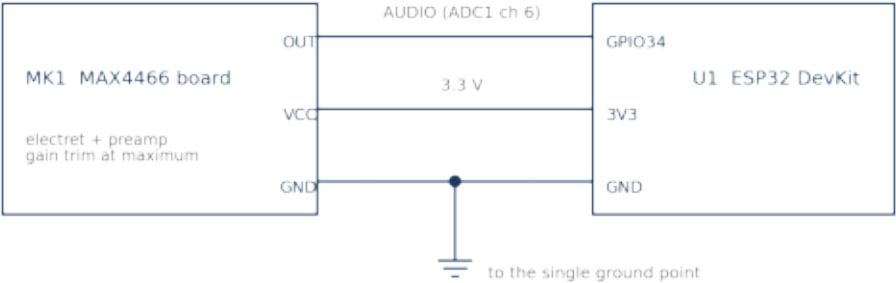
Audio Spectrum Visualizer — Schematic

Yuri (Iurii) Tiutvin

Rev 1 · 2026 · Sheet 1 of 1

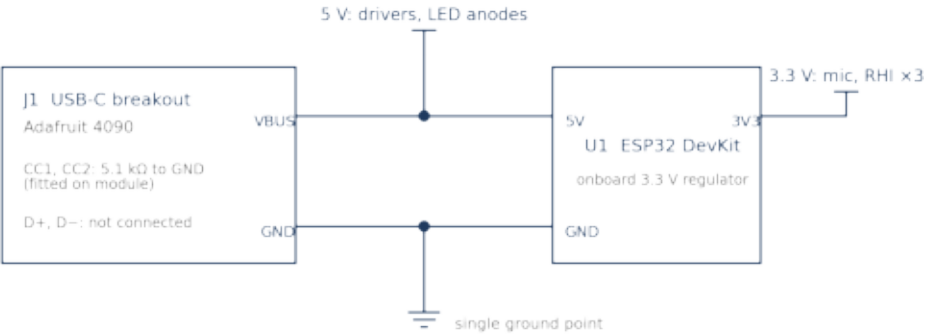
Drawn from the verified netlist in hardware/interconnect.md. Not a CAD export; there is no PCB project for this build.

1 · MICROPHONE INPUT



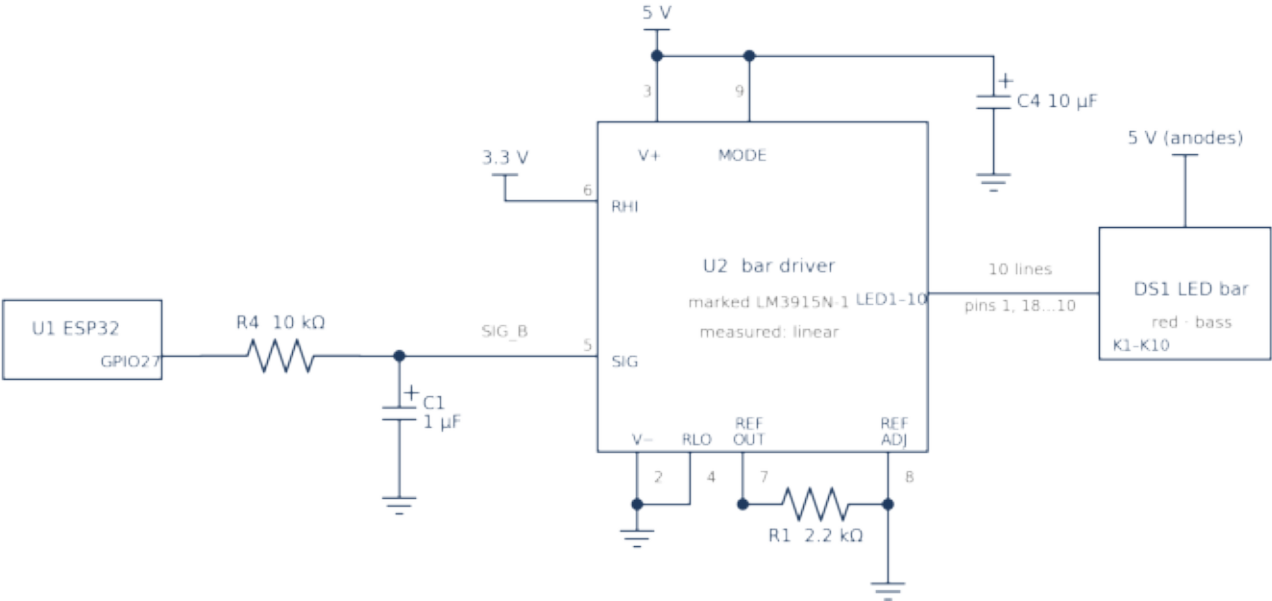
no coupling capacitor: the output idles at half of 3.3 V; the firmware subtracts it
measured on a same-model ESP32: quiet room ~1,940 counts = 1.66 V; loud music 600–3,290 counts

2 · POWER INPUT



measured: 5.178 V at VBUS before connecting · 70.7 mA with bars dark · 237.5 mA highest seen
never powered from the DevKit's own USB port at the same time

3 · DISPLAY CHANNEL — BASS SHOWN; MID AND TREBLE ARE IDENTICAL



measured on chips of the same part: LED 1 on at 0.31 V ... LED 10 at 3.26 V, mean step 0.328 V (linear, not the LM3915's 3 dB steps)
calculated: $I_{LED} = 12.5 \text{ V} / R1 = 5.7 \text{ mA}$; filter corner 15.9 Hz, ripple = 5 mV at 10 kHz

4 · CHANNELS

Band	GPIO	Filter	Driver	R1	Bar
Bass	27	R4, C1	U2	R1	DS1 red
Mid	26	R5, C2	U3	R2	DS2 yellow
Treble	25	R6, C3	U4	R3	DS3 green

Per channel: R4–R6 10 kΩ, C1–C3 1 μF,
R1–R3 2.2 kΩ, C4–C6 10 μF at each V+.
Bands: bass 80–300 Hz, mid 300 Hz–4 kHz,
treble 4–10 kHz (set in firmware).

One 3.3 V logic domain (ESP32, microphone, driver full-scale reference); drivers and LED anodes on 5 V from the USB-C input.
Driver/LED ground and microphone/ESP32 ground run on separate wires to one star point on the back of the back board.
ESP32 pins are GPIO numbers as printed on the DevKit; driver pins are 18-pin DIP numbers. Reference designators match hardware/bom.csv.